

BASIC ELECTRICAL ENGINEERING

UNIT – I

DC Circuits

1. Electrical Circuit Elements

- Resistor (R)
- Inductor (L)
- Capacitor (C)

2. Sources

- Voltage sources
- Current sources

3. Fundamental Laws

- Kirchhoff's Current Law (KCL)
- Kirchhoff's Voltage Law (KVL)

4. Analysis of DC Circuits

- Analysis of simple circuits with DC excitation
- Network Reduction Techniques

5. Circuit Analysis Methods

- Mesh Analysis
- Nodal Analysis

6. Network Theorems

- Superposition Theorem
- Thevenin's Theorem
- Norton's Theorem

UNIT – II

AC Circuits

1. Sinusoidal Waveforms

- Representation of sinusoidal waveforms
- Peak value
- RMS value

2. Phasor Representation

- Phasor diagrams
- Phasor analysis

3. Power in AC Circuits

- Real Power (Active Power)
- Reactive Power
- Apparent Power
- Power Factor

4. Single-Phase AC Circuits (Series Only)

- Pure R circuit
- Pure L circuit

- Pure C circuit
- RL series circuit
- RC series circuit
- RLC series circuit

5. Three-Phase Circuits

- Balanced three-phase circuits
- Star (Y) connection
 - Voltage relations
 - Current relations
- Delta (Δ) connection
 - Voltage relations
 - Current relations

UNIT – III

Transformers and Three-Phase Induction Motors

A. Transformers

1. Basic Principles

- Electromagnetic Induction
- Faraday's Laws of Electromagnetic Induction
- Lenz's Law
- Statically induced EMF

2. Magnetic Properties

- B-H Characteristics

3. Types of Transformers

- Ideal Transformer
- Practical Transformer

4. Losses and Efficiency

- Types of losses
- Efficiency calculation (Simple Problems only)

5. Three-Phase Transformer Connections

B. Three-Phase Induction Motor

1. Rotating Magnetic Field

• Generation of rotating magnetic field	UNIT – IV	5. Characteristics	UNIT – V
2. Construction and Working	Single-Phase Induction Motor and DC Machines	• Open Circuit Characteristic (OCC)	Electrical Installations
• Principle of operation • Constructional features		6. Applications	1. Components of LT Switchgear
3. Types	A. Single-Phase Induction Motor		• Switch Fuse Unit (SFU) • Miniature Circuit Breaker (MCB) • Earth Leakage Circuit Breaker (ELCB) • Moulded Case Circuit Breaker (MCCB)
• Squirrel Cage Induction Motor • Slip Ring Induction Motor	1. Construction	C. DC Motors	2. Wiring Materials
4. Applications	2. Principle of Operation	1. Principle of Operation	• Types of Wires • Types of Cables
	3. Types	2. Types of DC Motors	3. Earthing
	• Capacitor Start Motor • Capacitor Run Motor	3. Applications	• Importance of Earthing • Types of Earthing
	4. Applications		4. Basic Calculations
			• Energy consumption calculations • Power factor improvement
	B. DC Generators		
	1. Basic Principles		
	• Dynamically induced EMF • Fleming's Right-Hand Rule • Fleming's Left-Hand Rule		
	2. Construction and Operation		
	• Construction of DC Generator • Principle of operation		
	3. EMF Equation		
	4. Types of DC Generators		